Power of a Sine

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clc; clear; close all;
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Single Sine

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syms psi y n;
assume(n,{'positive','integer'});
syms ii;
ii = sqrt(-1);

sin_psi = (exp(ii*psi) - exp(-ii*psi))/(2*ii);
cos_psi = (exp(ii*psi) + exp(-ii*psi))/2;
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Single sine for powers $n = 1..4$

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expand(sin_psi^2)
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ans =

$$\frac{1}{2} - \frac{e^{2\psi i}}{4} - \frac{e^{-2\psi i}}{4}$$

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expand(sin_psi^3)
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ans =

$$-\frac{e^{-\psi i}\, e^{-2\psi i}\, i}{8} - \frac{3\, e^{-\psi i}\, e^{2\psi i}\, i}{8} + \frac{3\, e^{\psi i}\, e^{-2\psi i}\, i}{8} + \frac{e^{\psi i}\, e^{2\psi i}\, i}{8}$$

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expand(sin_psi^4)
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ans =

$$\frac{3}{8} - \frac{e^{2\psi i}}{4} + \frac{e^{-4\psi i}}{16} + \frac{e^{4\psi i}}{16} - \frac{e^{-2\psi i}}{4}$$

Single sine for odd powers

Start with the binomial theorem:

$$\left(\frac{e^{i\psi}}{2i} - \frac{e^{-i\psi}}{2i}\right)^{2n+1} = \sum_{k=0}^{2n+1} C_k^{2n+1} \left(\frac{e^{i\psi}}{2i}\right)^{2n+1-k} \left(-\frac{e^{-i\psi}}{2i}\right)^k$$

Multiply exponents in the numerator, factor out $(-1)^k$:

$$\left(\frac{e^{i\psi}}{2i} - \frac{e^{-i\psi}}{2i}\right)^{2n+1} = \sum_{k=0}^{2n+1} C_k^{2n+1} (-1)^k \left(\frac{e^{i(2n+1-k)\psi}}{(2i)^{2n+1-k}}\right) \left(\frac{e^{-i(k)\psi}}{(2i)^k}\right)$$

Multiply both fractions:

$$\left(\frac{e^{i\psi}}{2i}-\frac{e^{-i\psi}}{2i}\right)^{2n+1}=\sum_{k=0}^{2n+1}C_k^{2n+1}(-1)^k\frac{(e^{i(2n+1-k)\psi})(e^{-i(k)\psi})}{(2i)^{2n+1-k}(2i)^k}$$

Add exponents in the numerator and denominator:

$$\left(\frac{e^{i\psi}}{2i}-\frac{e^{-i\psi}}{2i}\right)^{2n+1}=\sum_{k=0}^{2n+1}C_k^{2n+1}(-1)^k\frac{(e^{i(2n+1-2k)\psi})}{(2i)^{2n+1}}$$

Distribute exponents in the denominator:

$$\left(\frac{e^{i\psi}}{2i}-\frac{e^{-i\psi}}{2i}\right)^{2n+1}=\sum_{k=0}^{2n+1}C_k^{2n+1}(-1)^k\frac{(e^{i(2n+1-2k)\psi})}{2^{2n+1}i^{2n+1}}$$

Factor out $\frac{1}{2^{2n+1}}$:

$$\left(\frac{e^{i\psi}}{2i}-\frac{e^{-i\psi}}{2i}\right)^{2n+1}=\frac{1}{2^{2n+1}}\sum_{k=0}^{2n+1}C_k^{2n+1}(-1)^k(e^{i(2n+1-2k)\psi})\frac{1}{i^{2n}i}$$

Move *i* to the numerator using $-i=\frac{1}{i}$:

$$\left(\frac{e^{i\psi}}{2i}-\frac{e^{-i\psi}}{2i}\right)^{2n+1}=\frac{1}{2^{2n+1}}\sum_{k=0}^{2n+1}C_k^{2n+1}(-1)^k(-i)e^{i(2n+1-2k)\psi}\frac{1}{i^{2n}}$$

Simplify $\frac{1}{i^{2n}}$ by applying $i^{2n}=(-1)^n$:

$$\left(\frac{e^{i\psi}}{2i}-\frac{e^{-i\psi}}{2i}\right)^{2n+1}=\frac{1}{2^{2n+1}}\sum_{k=0}^{2n+1}C_k^{2n+1}(-1)^k(-i)e^{i(2n+1-2k)\psi}\frac{1}{(-1)^n}$$

$$\left(\frac{e^{i\psi}}{2i}-\frac{e^{-i\psi}}{2i}\right)^{2n+1}=\frac{1}{2^{2n+1}}\sum_{k=0}^{2n+1}C_k^{2n+1}(-1)^{k-n}(-i)e^{i(2n+1-2k)\psi}$$

Break up the summation:

$$\left(\frac{e^{i\psi}}{2i}-\frac{e^{-i\psi}}{2i}\right)^{2n+1}=\frac{1}{2^{2n+1}}\left\{\sum_{k=0}^{n}C_k^{2n+1}(-1)^{k-n}(-i)e^{i(2n+1-2k)\psi}+\sum_{k=n+1}^{2n+1}C_k^{2n+1}(-1)^{k-n}(-i)e^{i(2n+1-2k)\psi}\right\}$$

Set the bounds of the second sum to $k=0..n$ by applying $k\to 2n+1-k$

$$\left(\frac{e^{i\psi}}{2i}-\frac{e^{-i\psi}}{2i}\right)^{2n+1}=\frac{1}{2^{2n+1}}\left\{\sum_{k=0}^{n}C_k^{2n+1}(-1)^{k-n}(-i)e^{i(2n+1-2k)\psi}+\sum_{k=0}^{n}C_{2n+1-k}^{2n+1}(-1)^{2n+1-k-n}(-i)e^{i(2n+1-2(2n+1-k))\psi}\right\}$$

Apply the symetry rule for binomial coefficients:

$$\left(\frac{e^{i\psi}}{2i}-\frac{e^{-i\psi}}{2i}\right)^{2n+1}=\frac{1}{2^{2n+1}}\left\{\sum_{k=0}^{n}C_k^{2n+1}(-1)^{k-n}(-i)e^{i(2n+1-2k)\psi}+\sum_{k=0}^{n}C_k^{2n+1}(-1)^{2n+1-k-n}(-i)e^{i(2n+1-2(2n+1-k))\psi}\right\}$$

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syms n k
simplify(2*n+1-2*(2*n+1-k))
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ans = $2\,k - 2\,n - 1$

Note that $2n + 1 - 2(2n + 1 - k) = 2k - 2n - 1 = -(2n + 1 - 2k)$

$$\left(\frac{e^{i\psi}}{2i} - \frac{e^{-i\psi}}{2i}\right)^{2n+1} = \frac{1}{2^{2n+1}}\left\{\sum_{k=0}^{n} C_k^{2n+1}(-1)^{k-n}(-i)e^{i\,(2n+1-2k)\,\psi} + \sum_{k=0}^{n} C_k^{2n+1}(-1)^{2n+1-k-n}(-i)e^{-i\,(2n+1-2k)\,\psi}\right\}$$

Note that $(-1)^{2n} = 1, (-1)^1 = -1$

$$\left(\frac{e^{i\psi}}{2i} - \frac{e^{-i\psi}}{2i}\right)^{2n+1} = \frac{1}{2^{2n+1}}\left\{\sum_{k=0}^{n} C_k^{2n+1}(-1)^{k-n}(-i)e^{i\,(2n+1-2k)\,\psi} - \sum_{k=0}^{n} C_k^{2n+1}(-1)^{-k-n}(-i)e^{-i\,(2n+1-2k)\,\psi}\right\}$$

Note that $(-1)^k = (-1)^{-k}$

$$\left(\frac{e^{i\psi}}{2i} - \frac{e^{-i\psi}}{2i}\right)^{2n+1} = \frac{1}{2^{2n+1}}\left\{\sum_{k=0}^{n} C_k^{2n+1}(-1)^{k-n}(-i)e^{i\,(2n+1-2k)\,\psi} - \sum_{k=0}^{n} C_k^{2n+1}(-1)^{k-n}(-i)e^{-i\,(2n+1-2k)\,\psi}\right\}$$

Recombine the sums:

$$\left(\frac{e^{i\psi}}{2i} - \frac{e^{-i\psi}}{2i}\right)^{2n+1} = \frac{1}{2^{2n+1}}\sum_{k=0}^{n} C_k^{2n+1}(-1)^{k-n}(-i)\left\{e^{i\,(2n+1-2k)\,\psi} - e^{-i\,(2n+1-2k)\,\psi}\right\}$$

Note that: $\sin(u) = \dfrac{e^{i\,u} - e^{-i\,u}}{2i}$

And therefore $2\sin(u) = (-i)e^{i\,u} - (-i)e^{-i\,u}$

$$\left(\frac{e^{i\psi}}{2i} - \frac{e^{-i\psi}}{2i}\right)^{2n+1} = \frac{2}{2^{2n+1}}\sum_{k=0}^{n} C_k^{2n+1}(-1)^{k-n}\sin(2n+1-k)\psi$$

$$\left(\frac{e^{i\psi}}{2i} - \frac{e^{-i\psi}}{2i}\right)^{2n+1} = \frac{1}{2^{2n}}\sum_{k=0}^{n} C_k^{2n+1}(-1)^{k-n}\sin(2n+1-k)\psi$$

Factor out the $(-1)^{-n}$ and reapply euler's formula on the left hand side. Again take agvantage of $(-1)^n = (-1)^{-n}$.

$$\sin^{2n+1}\psi = \frac{(-1)^n}{2^{2n}}\sum_{k=0}^{n} C_k^{2n+1}(-1)^k\sin(2n+1-k)\psi$$